A droplet-based microfluidic screening system for the discovery of slow-growing and novel microbes in environmental samples

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환경 시료 내 느리게 성장하는 신규 미생물 발굴을 위한 액적 기반 미세유체 스크리닝 시스템

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Abstract

The discovery of new microbes and establishment of microbial libraries are essential for advancing ecological and industrial research. Conventional isolation methods, such as agar plate cultures, often fail to identify new microbes, particularly slow-growing species. Here, we present an automated droplet microfluidic screening system designed to overcome the challenge by enabling long-term culture and high-throughput analysis of environmental microbes with single-cell resolution. The system utilizes fluorescence-based detection within droplets, combined with iterative culture and sorting processes. This systems identified 6,539 new microbes over four weeks, significantly increasing microbial diversity and unclassified variants. Genetic analysis and interaction studies with harmful algae demonstrated potential applications. This approach offers a transformative platform for discovering slow-growing microbes and exploring microbial diversity in environmental.

Keywords: ***Droplet Microfluidics (액적 미세유체), High-throughput Screening (고성능 스크리닝), Microbes Discovery(미생물 발굴), Long-Term Culture(장기배양)***

1. Introduction

The exploration of new microbes and the establishment of comprehensive microbial libraries are fundamental to expanding our understanding of ecological diversity and setting the stage for progress in various industrial and pharmaceutical fields [1]. Conventionally, isolation methods using agar plates are widely used, however limited in discovering new microbes, particularly slow-growing microbes, requiring long-term culture. Typically, the agar plates dry out during long-term culture [2], and slow-growing bacteria are also suppressed by already identified, faster-growing species [3]. Thus, new microbe isolation method enabling long-term culture needs to be investigated. Here, we introduce a droplet microfluidic screening system capable of effectively discovering new microbes, including both fast- and slow-growing ones, from the environment.

2. Experimental Section

2.1. Device design and screening process

The system consists of two different devices; droplet generation/culture device that produces and cultures droplets containing single microbe, and droplet detection/sorting device that analyzes microbial growth within droplets and sort particular droplets showing higher growth (Fig.1). The long-term culture and isolation of slow-growing microbes are achieved through weekly iterative screening process (Fig.2); Environmental microbes and growth-tracking fluorescent dye (alamarBlue) are encapsulated as single cells within droplets, and cultured in the downstream chamber (Fig.1a). Alamarblue is utilized to analyze the growth of unknown microbes, characterized by measuring the fluorescent intensity of droplets, where more cell growth results in higher intensity. After one week of culture, the droplets are transferred to the droplet detection/sorting device, and the microbial growth of each droplet is analyzed (Fig.1b). Droplets with fast-growing microbes are extracted for further off-chip analysis/storage, while low-signal droplets (slow-growing microbes) move back to droplet generation/culture device for additional culture. The culture-detection/sorting cycle is repeated on the recollected droplets at weekly intervals for 4~6 weeks. By repeating the culture-detection/sorting process on the identical sample, the developed system can provide sufficient long-term culturing period for slow-growing cells to grow and classify environmental microbes based on growth rates.

2.2. Environmental sample screening

Environmental microbes collected from Daecheong Lake (South Korea) were applied to developed screening system for discovering new microbes (Fig.3). Approximately 36,000 droplets with single microbe were screened for 4 weeks, and 6,539 new microbes were discovered. Interestingly, the percentage of “Others”

(unknown/new microbes) becomes higher as screening step increases (Fig.3b). In addition, the number of unassigned microbial ASVs (Amplicon Sequence Variant) increases with more screening steps, indicating the increasing number of new microbes, not classified at gene level, and higher level of microbial diversity (Fig.3c).

2.3. Environmental sample screening

Genetic analysis and microbial interaction study from newly discovered microbes are being conducted. Microcystis, harmful microalga related to algal blooms, was co-cultured with new microbes, and their interaction was analyzed (Fig.6), where Microcystis death caused by microbial interaction was confirmed by both droplet and flask cultures.

3. Conclusion

This study presents a droplet-based microfluidic screening system that overcomes the limitations of conventional microbial isolation methods, enabling the discovery of slow-growing microbes from environmental samples. The proposed approach provides an effective platform for exploring microbial diversity and supports broader applications in environmental and biological screening.

Acknowledgments

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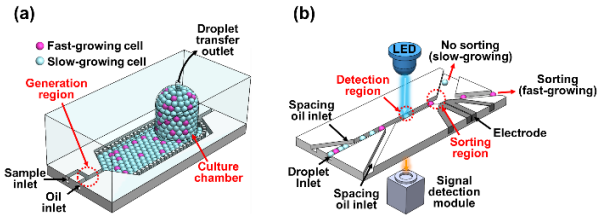

Figure 1. Illustration of (a) a droplet generation/culture device and (b) a droplet detection/sorting device

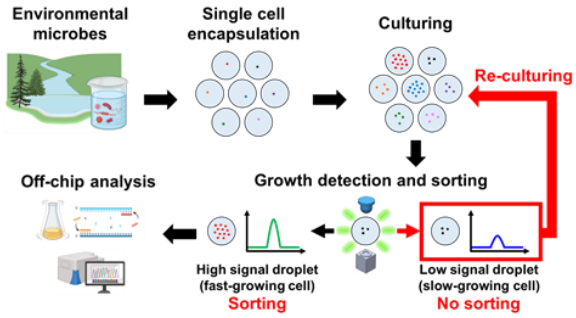

Figure 2. A schematic showing the screening process to discover new microbes based on their growth. The iterative screening process allows for the long-term droplet culture and discovery of slow-growing microbes.

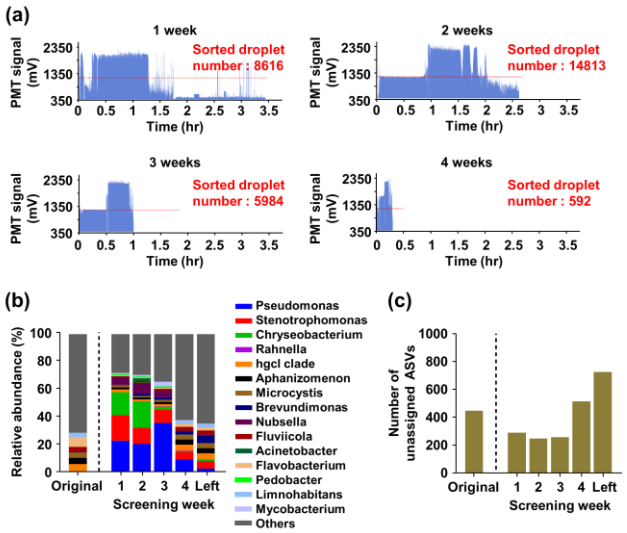

Figure 3. Environmental sample screening results. (a) Weekly droplet growth detection and sorting over 4-week screening period. (b) Relative abundance of well-grown microbial taxa at the genus level. The percentage of Others (unknown, new microbes) increases as the screening steps increases. (c) The number of unassigned microbial ASVs (Amplicon Sequence Variant). As the screening steps increase, higher ASVs was observed, implying higher level of diversity and unknown microbe species.

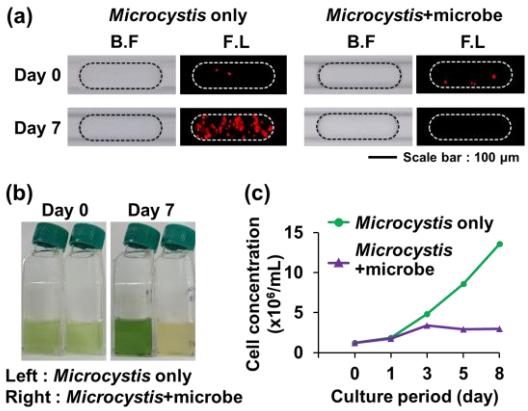

Figure 4. Interaction analysis between a newly discovered microbe and Microcystis. (a) Cell growth comparison with and without microbial interaction. The initial cell number of Microcystis in droplet is 5. Microbe-microalgae interaction analysis in flask scale. (b) Photography of flask culture and (c) Growth analysis with and without microbial interaction, monitored by cell-counting.